

RAG Advantages and Limitations

RAG provides substantial advantages over traditional large language models alone. By grounding responses in retrieved documents, RAG significantly reduces hallucinations and factual errors, producing more reliable and trustworthy outputs. Systems can be updated dynamically by modifying the knowledge base without retraining the generative model, enabling rapid adaptation to new information. Retrieved documents serve as citations, enabling users to verify information sources and understand response justifications. RAG extends the effective knowledge of language models beyond their training data, accessing real-time information and proprietary documents. The hybrid approach is more cost-effective than fine-tuning large models, requiring only infrastructure for storage and retrieval. RAG enables better control over which information influences responses, improving security and compliance in sensitive domains. However, RAG has limitations. Retrieval quality directly impacts generation quality—poor retrieval results in poor responses regardless of generator capability. The system depends on high-quality, well-organized knowledge bases; garbage in produces garbage out. Retrieval-generation misalignment can occur when relevant documents don't contain needed information or when context window size limits how much information can be included. Computational overhead from retrieval adds latency compared to direct generation. Multi-hop reasoning requiring information synthesis across multiple documents remains challenging. Maintaining, updating, and scaling knowledge bases demands significant operational effort.